Law of Sines

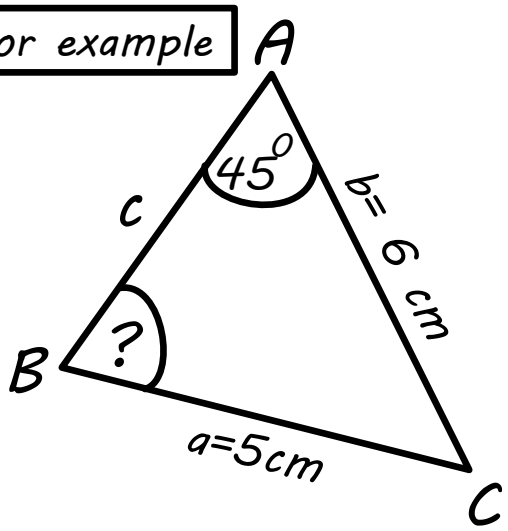
Finding **angles**

To use the law of sines we need to know…

2 sides

1 angle which is opposite to one of these sides

For example



Now we can use the sine rule:

$$\frac{\sin A}{a} = \frac{\sin B}{b} = \frac{\sin C}{c}$$

what we know…

a=5cm	A=45°
b=6cm	B= ?°

For this example··

$$\frac{\sin 45}{5} = \frac{\sin B}{6}$$

$$\sin B = \frac{6\sin 45}{5}$$

$$\sin B = 0{\cdot}8485\cdots$$

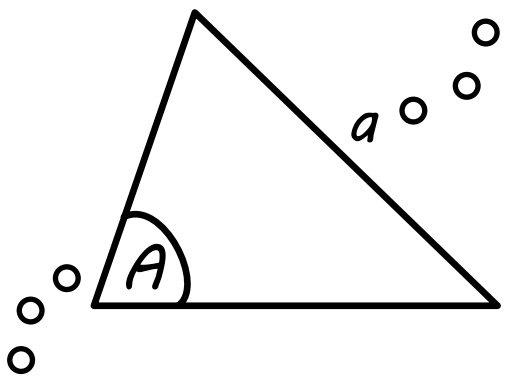
$$B = \sin^{-1} 0{\cdot}8485\cdots$$

$$B = 58{\cdot}0519\cdots$$

B = _____°(to 1 d·p)

Labelling triangles

If this side is "a"



The opposite angle is "A"

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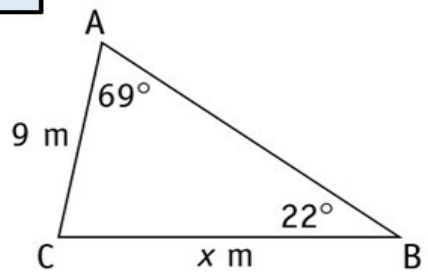
Lesson Objective: By the end of the lesson you should be very comfortable with using the Sine Rule and when to use it as well as how to rearrange when needed.

1



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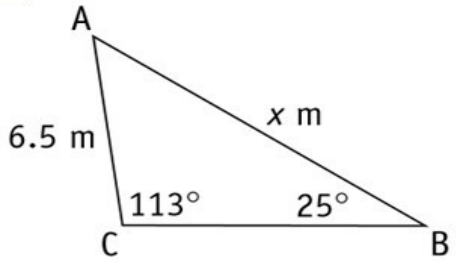

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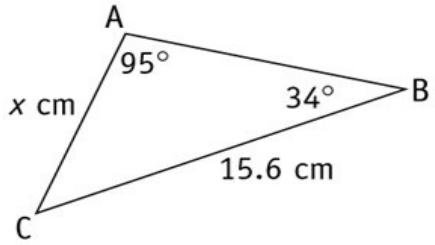

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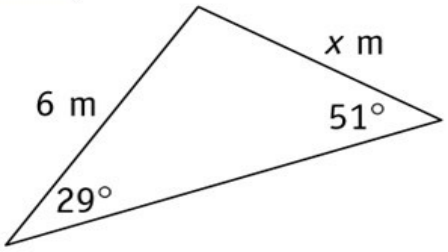

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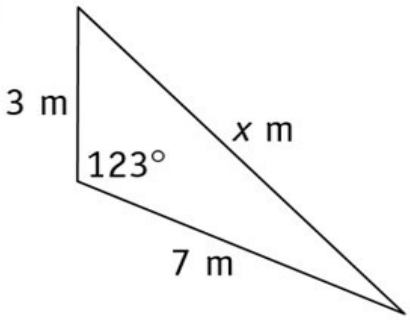

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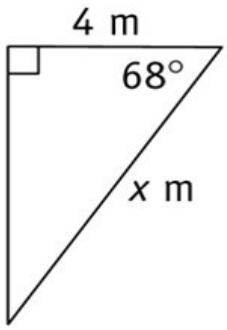

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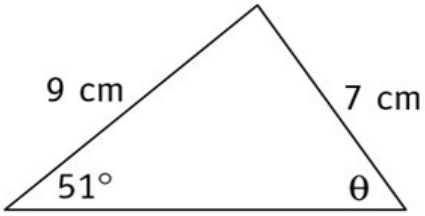

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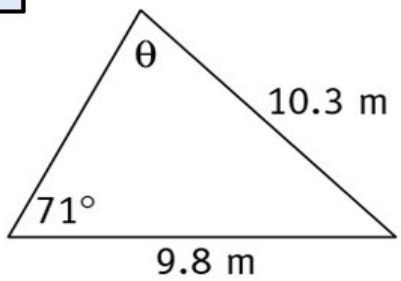

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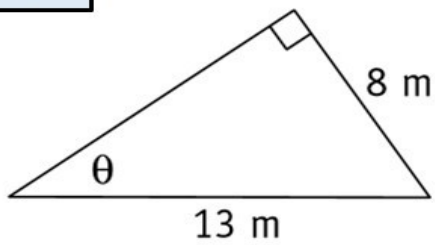

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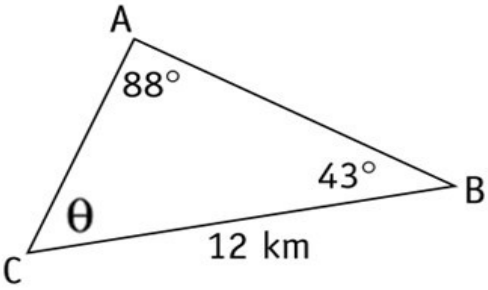

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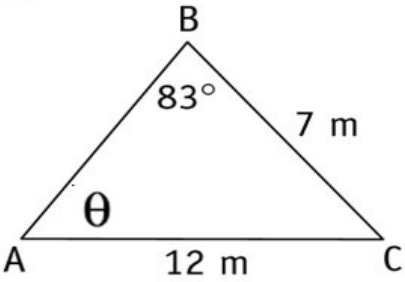

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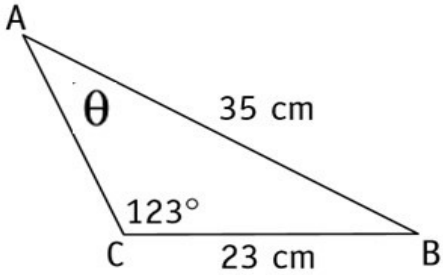

Self Evaluation: What have you learnt today? Are you confident with the sine rule? Did you struggle when finding the angle instead of a side? Or with rearranging?